

# Types, Variables and Input

## L10 - Variable-length types

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- An algorithm is a **procedure used for solving a problem or performing a computation.**
- Algorithmic thinking, then, is the construction of an algorithm, or step-by-step process for solving a problem and similar problems.

- Algorithmic thinking is the foundation of computer science and the process to develop code and program applications.
- In other words, algorithmic thinking is not solving for a specific answer; instead, it solves how to build a sequential, complete, and replicable process that has an end point – an algorithm.

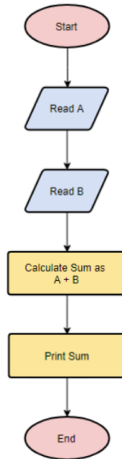
## Examples of scenarios to apply Algorithmic Thinking

- Boiling a kettle
- Escaping the Canyon
- Tying shoelaces
- Cooking your favourite meal

## Flow Charts

- A flowchart is a visual representation of an algorithm, typically using symbols and arrows to depict the sequence of steps involved.
- This visual representation enables both programmers and non-programmers to understand the logic behind the algorithm.

# Flow Chart Example



# Program Structure

- C++ is an object oriented programming language that was developed as an extension of the C programming language.
- C++ provides a rich standard library that includes a wide range of functions and classes for various tasks, such as input/output operations, data structures, algorithms, and more.

## Example of C++ code

```
1 #include <iostream>
2
3 using namespace std;
4
5 int main() {
6     //Your code here
7     return 0;
8 }
```

## Important parts of a C++ program

- keywords
- compiler directive
- namespace
- main function
- statement
  - block statement
  - return statement
- comments

## Data types used in C++

- Primitive (or Built-in) Types
- Derived Types
- Abstract (or User-defined) Types

# Primitive Data Types

- These data types are **built-in or predefined data types and can be used directly by the user to declare** variables.
- They are directly supported by the compiler and typically have predefined operations associated with them.
- Examples include **integer types** (int, short, long), **floating-point types** (float, double, long double), **character types** (char), and **boolean type** (bool).

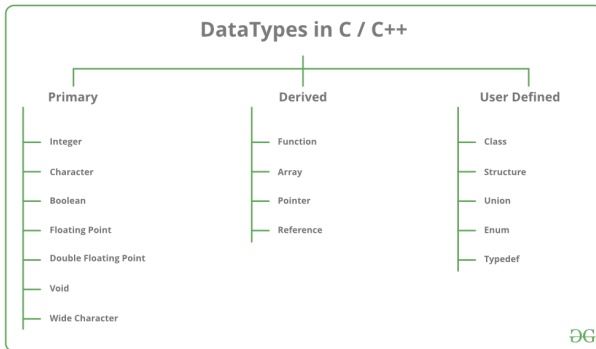


Figure: Data Types in C++

<https://www.geeksforgeeks.org/cpp-data-types/>

## Various Primitive Data Types

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int number = 20;
6     float myfloat= 10.11;
7     double mydecimal = 100.3454;
8     bool mybool1 = true;
9     bool mybool2 = 0;
10    char mychar = 'C';
11    string mystring = "Welcome to COS-132";
12
13    cout << "My-integer:-" << number << endl;
14    cout << "My-Float:-" << myfloat << endl;
15    cout << "My-Decimal:-" << mydecimal << endl;
16    cout << "Boolean-state:-" << mybool1 << "or-" << mybool2 << endl;
17    cout << "My-character:-" << mychar << endl;
18    cout << "The-string-value-is:-" << mystring << endl;
19    return 0;
20 }
```

## 1. Ordinal

Ordinal types are Integer, Character and Boolean. The keywords used to represent these types are `int`, `char` and `bool` respectively.

## 2. Variable-length

Variable-length types include floating point, of which there are two types: `float` and `double`. The difference lies in precision.

Floating point data type stores single-precision floating-point values, with float typically requiring 4 bytes of memory space.

Double floating point, on the other hand, stores double-precision floating-point values and typically requires 8 bytes of memory space.

| Type                               | C++ type name      | Byte size | Range                                                    |
|------------------------------------|--------------------|-----------|----------------------------------------------------------|
| Ordinal                            | int                | 4         | -2147483648 $\rightarrow$ 2147483647                     |
|                                    | char               | 1         | -128 $\rightarrow$ 127                                   |
|                                    | bool               | 1         | true or false                                            |
| Variable-length                    | float              | 4         | $-3.4 \times 10^{38} \rightarrow 3.4 \times 10^{38}$     |
|                                    | double             | 8         | $-1.7 \times 10^{308} \rightarrow 1.7 \times 10^{308}$   |
| Variable-length<br>(Library-based) | std::string        | 24        |                                                          |
| Modifiers                          | short int          | 2         | -32768 $\rightarrow$ 32767                               |
|                                    | unsigned short int | 2         | 0 $\rightarrow$ 65535                                    |
|                                    | unsigned long int  | 4         | 0 $\rightarrow$ 4294967295                               |
|                                    | long long int      | 8         | $-2^{63} \rightarrow 2^{63}-1$                           |
|                                    | long double        | 12        | $-1.1 \times 10^{4932} \rightarrow 1.1 \times 10^{4932}$ |

Figure: Basic and Modified Types

Write the C++ program for the Flow chart example showed in this lecture.

## Example Solution

```
1 #include <iostream>
2
3 using namespace std;
4 int main()
5 {
6     //declare variables
7     int a;
8     int b;
9     //input numbers
10    cout<< "Enter first number:-";
11    cin>> a;
12    cout<< "Enter second number:-";
13    cin>>b;
14    //calculate the sum
15    cout<< "The sum is:-" << a + b <<endl;
16    return 0;
17 }
```

Cyril traveled by car from Pretoria to Bloemfontein. It took him four hours to travel between the two cities. Write a C++ program that will calculate the speed at which he was traveling. You are told the distance between the two cities is 450 km.

## Example Solution

```
1 #include <iostream>
2 using namespace std;
3 int main() {
4     // declare variables
5     double mydistance = 450;
6     double mytime = 4;
7     // Calculate speed
8     double speed = mydistance / mytime;
9     // Output the result
10    cout << "Cyril's speed:-" << speed << "-km/h" << endl;
11    return 0;
12 }
```

- Modify the above code such that it takes in the distance and time from the user.